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Evolution of the structure factor in a hyperbolic model of spinodal decomposition

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Abstract. We consider the modification of the Cahn-Hilliard equation when a time delay process through a memory function is taken into account. The memory effects are seen to affect the dynamics of phase transition at short times. The process of fast spinodal decomposition associated with a conserved order parameter - concentration is studied numerically. Details of a semi-implicit numerical scheme used to simulate the kinetics of spinodal decomposition and evolution of the structure factor are discussed. Analysis of the modeled structure factor predicted by a hyperbolic model of spinodal decomposition is presented in comparison with the parabolic model of Cahn and Hilliard. It is shown that during initial periods of decomposition the structure factor exhibits wave behavior. Analytical treatments explain such behavior by existence of damped oscillations in structure factor at earliest stages of phase separation and at large values of the wave-number. These oscillations disappear gradually in time and the hyperbolic evolution approaches the pure dissipative parabolic evolution of spinodal decomposition.

1 Introduction

Spinodal decomposition is a well-known phenomenon in which, as a result of spontaneous growth of fluctuations, both phases have equivalent symmetry but differ only in composition. It has been theoretically described within linear theories [1,2] and statistical theory with non-linear atomic mobility [3], which were intensively tested against experimental data [4–6]. Overviews on these studies are given in Refs. [7–9].

Most studies of binary alloys, oxides or glasses that were deeply quenched into the miscibility gap failed to corroborate the predictions of the linear theory of Cahn and Hilliard (CH) [1] or the linear theory with noise given by the Cahn-Hilliard-Cook (CHC) model [2]. It can be seen in functions characterizing patterns evolving during spinodal decomposition. In particular, one of the non-linear functions which is extracted from experimental data (specifically, from the structure factor) is the amplification rate of decomposition *versus* square of the wave-number [4–6]. Therefore, to describe non-linear effects in spinodal decomposition several modifications of the linear theory were carried out. To describe the non-linearity arising in the amplification rate the LBM statistical model (Langer, Bar-on and Miller model) [3] has been intensively tested. This model yields two coupled equations of motion: one for the structure factor and one for the one-point probability density function. In comparison to the classical linear models [1,2], the LBM-model contains an additional contribution having a meaning similar to a diffusion term. This contribution leads to non-linearities in the amplification rate of decomposition (see Fig. 4 in Ref. [3]) similar to those extracted from experimental data [4–6].

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Direct quantitative comparison of the LBM-model with experimental data on spinodal decomposition in binary alloys leads to the conclusions that (i) one of the features predicted by the model, namely, a parallel tangent for the amplification rate at a maximum wave-number, was not observed [5], and (ii) the atomic mobility must depend on composition [6].

Predictions of the LBM-model were also compared with data on the time evolution of the structure factor for a spinodally decomposing binary liquid. Using precise light-scattering measurements of early-stage spinodal decomposition in a near-critical binary liquid mixture obtained by Bailey [10], an extended version of the LBM-model was adapted to the real experimental conditions by Donley [11]. As it follows from results of Donley and Langer [12], using the LBM-model, the computed structure factor disagrees with experimental data during the earliest stages of decomposition but merges closely with experiment in the following stages (see Fig. 2 in Ref. [12]).

These disagreements might be attributed to the strong influence of local nonequilibrium in the diffusion field of a spinodally decomposing binary system, especially during the initial stages of decomposition [13,14]. Introduction of the new contribution responsible for the purely non-equilibrium part of entropy (or free energy) leads to a non-linearity in the amplification rate for decomposition [14] and to good agreement with experimental data, at least for a spinodally decomposed binary glass [15]. This contribution changes a model based on a parabolic partial differential equation [1–3] to a model with memory [16] able to describe both fast decomposition existing at the early stages and slow spinodal decomposition occurring at the following and final stages. In particular, the memory function for the diffusion flux in a form of exponential decay leads to a "hyperbolic model" described by a partial differential equation of the hyperbolic type.

In the present Article, a description of the structure factor which directly characterizes patterns in spinodally decomposing system is given. We present numerical results for the evolution of the structure factor and analytically explain differences arising in its behavior predicted by the hyperbolic and parabolic models of separating phases, particularly, at the earliest stages of spinodal decomposition.

2 Governing equations for the hyperbolic model

We consider a binary system free of imperfections and neglect the effect of coherency strain (for crystalline or amorphous solids [1,17,18]), anisotropy (as in a blend of liquid crystals or a polymer solution [19]), and shear flow (in liquids [20]) which may occur in a phase separation. In this case, the process of phase separation is controlled by atomic diffusion.

Whole stages of the dynamics of the transition from unstable to metastable or even stable phase state can be described through introduction of the prehistory of the system evolution [16]. To take the memory effect into account, the current for the concentration $c(\mathbf{r}, t)$ of a second chemical component in the binary system can be written as:

$$\mathbf{j} = -\int_{-\infty}^{t} M_R(t - t^*)\nabla\left(\frac{\delta\mathcal{F}(t^*, \mathbf{r})}{\delta c}\right) dt^*, \tag{1}$$

with $M_R(t - t^*)$ the memory function which adopts the correlation between the system's dynamics at the past moment of time t^* and the current dynamics at the time t. Together with the balance law $\partial c/\partial t = -\nabla \cdot \mathbf{j}$, Eq. (1) gives the following equation for the arbitrary, slow or fast, dynamics:

$$\frac{\partial c}{\partial t} = \nabla \cdot \int_{-\infty}^{t} M_R(t - t^*)\nabla\left(\frac{\delta\mathcal{F}(t^*, \mathbf{r})}{\delta c}\right) dt^*. \tag{2}$$

Taking the free energy $\mathcal{F}$ in the general Ginzburg-Landau form:

$$\mathcal{F}[c(t^*, \mathbf{r})] = \int_{v_0}\left(f(T, c) + \frac{\epsilon_c^2}{2}|\nabla c|^2\right) dv_0, \tag{3}$$

and the memory function of the exponential form $M_R(t-t^*) = (M/\tau_D)\exp[-(t-t^*)/\tau_D]$, Eq. (2) can be rewritten as [13]

$$\tau_D \frac{\partial^2 c}{\partial^2 t} + \frac{\partial c}{\partial t} = \nabla \cdot \left[M \nabla \left(f'_c - \epsilon_c^2 \nabla^2 c \right) \right], \tag{4}$$

where v_0 is the subvolume of the system, τ_D the time for relaxation of the diffusion flux $\mathbf{j}$ to its steady-state, M the atomic mobility, ϵ_c^2 the parameter characterizing the length between the phase interfaces (which is proportional to the surface energy), f'_c the first derivative of the free energy density $f(T,c)$ with respect to concentration c, and T is the temperature. With $\tau_D \to 0$, Eq. (4) transforms into the Cahn-Hilliard equation [1]. In the present form, Eq. (4) can be considered as a modified Cahn-Hilliard equation which is a partial differential equation of hyperbolic type with the term $\tau_D \partial^2 c/\partial^2 t$ representing the delay in the separation of phases.

A natural boundary condition, following from the variation of functional (3), is given by

$$\epsilon_c^2 \frac{\partial c}{\partial t} \nabla_n c = \left(f'_c - \epsilon_c^2 \nabla^2 c \right) j_n, \tag{5}$$

where the sub-index n indicates a vector projection on the normal vector to the outer boundary of the system. Obviously, simple boundary conditions which satisfy Eq. (5) are described by

$$c \equiv const \qquad and \qquad \nabla c = 0. \tag{6}$$

These conditions must hold at any point on the outer surface around the volume v_0.

3 Numerical method

Let us assume the mobility M to be independent of position (independent of concentration) in Eq. (4). To obtain a normalized constitutive equation, the following dimensionless variables and parameters are introduced: $\tilde{t} = tM\Delta f/l_D^2$ the dimensionless time, $\tilde{\nabla} = \nabla/l_D$ the normalized operator "nabla", $\tilde{f}'_c \equiv \partial \tilde{f}/\partial c = f'_c/\Delta f$ the dimensionless characteristic for the free energy density, $\tilde{\epsilon}_c^2 = \epsilon_c^2 l_D^2/\Delta f$ the dimensionless parameter for the length scale of the interface which determines the relative contribution of the surface energy, l_D the spatial length for diffusion, Δf the characteristic scale (barrier height) for the free energy density and $\tilde{\tau}_D = \tau_D M \Delta f/l_D^2$ the dimensionless relaxation time. Then, Eq. (4) to be numerically solved takes the form:

$$\frac{\partial^2 c}{\partial^2 \tilde{t}} + \frac{1}{\tilde{\tau}_D}\frac{\partial c}{\partial \tilde{t}} = \frac{1}{\tilde{\tau}_D}\left[\tilde{\nabla}^2 \tilde{f}'_c - \tilde{\epsilon}_c^2 \tilde{\nabla}^4 c \right]. \tag{7}$$

From this it follows that the two relevant dimensionless parameters $\tilde{\tau}_D$ and $\tilde{\epsilon}_c^2$ characterize the decomposition and the region of computational stability.

The finite difference approximation c^n at the time t_n used in the treatment of Eq. (7) gives

$$\frac{1}{\Delta t}\left\{ \delta c^n - \delta c^{n-1} \right\} + \tilde{\tau}_D^{-1} \delta c^n = \tilde{\tau}_D^{-1} \tilde{\nabla}^2 \left(\tilde{f}'_c(c^{n-1}) - \tilde{\epsilon}_c^2 \tilde{\nabla}^2 c^n \right). \tag{8}$$

The fourth order derivative in position can either be taken at the time $\tilde{t}_n$ or $\tilde{t}_{n-1}$.

For equations of parabolic type, Chen and Shen [21] have shown that the semi-implicit scheme for the Time-Dependent-Ginzburg-Landau and Cahn-Hilliard equations is accurate and stable. The most important advantage is that this method is about 500 times more efficient than the explicit Euler finite-difference scheme in 3D. Therefore, an applicability of the semi-implicit scheme is also expected to be efficient for the hyperbolic Eq. (8).

To formulate a semi-implicit expansion for Eq. (8), it is preferable to take this last term at the new time $\tilde{t}_n$. Therefore, expanding the terms δc^n and δc^{n-1} in Eq. (8), one gets

$$\frac{1}{\Delta t}\left\{ \frac{c^n - c^{n-1}}{\Delta t} - \frac{c^{n-1} - c^{n-2}}{\Delta t} \right\} + \tilde{\tau}_D^{-1} \frac{c^n - c^{n-1}}{\Delta t} = \tilde{\tau}_D^{-1} \tilde{\nabla}^2 \left(\tilde{f}'_c(c^{n-1}) - \tilde{\epsilon}_c^2 \tilde{\nabla}^2 c^n \right). \tag{9}$$

Using the Fourier-spectral method for the spatial variables and treating the linear fourth-order operators implicitly and the non-linear terms explicitly, the first-order semi-implicit Fourier-spectral scheme is described by

$$\frac{1}{\Delta t}\left\{\frac{\hat{c}_k^n-\hat{c}_k^{n-1}}{\Delta t}-\frac{\hat{c}_k^{n-1}-\hat{c}_k^{n-2}}{\Delta t}\right\}+\tilde{\tau}_D^{-1}\frac{\hat{c}_k^n-\hat{c}_k^{n-1}}{\Delta t}=-\hat{k}^2\tilde{\tau}_D^{-1}\left(\left(\widehat{\tilde{f}_c'(c^{n-1})}\right)_k+\hat{k}^2\tilde{\epsilon}_c^2\hat{c}^n\right). \quad (10)$$

Here $\hat{c}_k^n$ is the Fourier transform of c^n, $\left(\widehat{\tilde{f}_c'(c^{n-1})}\right)_k$ is the Fourier transform of the non-linear term $\tilde{f}_c'(c^{n-1})$, $\hat{\mathbf{k}}=(k_1,k_2,k_3)$ is a vector in the Fourier space, and $\hat{k}=\sqrt{k_1^2+k_2^2+k_3^2}$ is the magnitude of $\hat{\mathbf{k}}$.

Collecting the terms in Eq. (10)

$$\hat{c}_k^n\left(\frac{1}{\Delta t^2}+\frac{\tilde{\tau}_D^{-1}}{\Delta t}+\hat{k}^4\tilde{\tau}_D^{-1}\epsilon_c^2\right)-\hat{c}_k^{n-1}\left(\frac{2}{\Delta t^2}+\frac{\tilde{\tau}_D^{-1}}{\Delta t}\right)+\frac{1}{\Delta t^2}\hat{c}_k^{n-2}=-\hat{k}^2\tilde{\tau}_D^{-1}\left(\widehat{\tilde{f}_c'(c^{n-1})}\right)_k,$$

finally, one gets

$$\hat{c}_k^n=\frac{\hat{c}_k^{n-1}\left[2(\Delta t)^{-2}-(\tilde{\tau}_D\Delta t)^{-1}\right]-(\Delta t)^{-2}\hat{c}_k^{n-2}-\hat{k}^2\tilde{\tau}_D^{-1}\left(\widehat{\tilde{f}_c'(c^{n-1})}\right)_k}{(\Delta t)^{-2}+(\tilde{\tau}_D\Delta t)^{-1}+\hat{k}^4\tilde{\tau}_D^{-1}\tilde{\epsilon}_c^2}. \quad (11)$$

Eq. (11) represents the semi-implicit numerical scheme to solving the hyperbolic equation. It allows to obtain the concentration $\hat{c}_k^n$ at the current time step n by previously established values $\hat{c}_k^{n-1}$ and $\hat{c}_k^{n-2}$ at the time steps $n-1$ and $n-2$, respectively.

To obtain results in numerical simulation, periodic boundary conditions satisfying Eq. (6) and the initial conditions $c(\mathbf{r},0)=c_0+\delta c$, $\partial c(\mathbf{r},0)/\partial t=0$ were used with c_0 the initial (nominal) concentration of the binary system and $\delta c|_{t=0}$ a Langevin noise term.

From Eq. (11), it is seen that two parameters are responsible for the stability and behavior of the spinodal decomposition characteristics: $\tilde{\tau}_D$ which corresponds to the relaxation time and $\tilde{\epsilon}_c^2$ which is a measure of the interfacial energy. As a result, modeling has been made in the following region of parameters $0.1\le\tilde{\tau}_D\le 2.0$ and $0.01\le\tilde{\epsilon}_c^2\le 1.0$ giving us computational stability in solution of Eq. (11).

Modeling has been made in a computational domain with the size of 256^3 grid nodes and uniform grid size of $\Delta x=1.0$. Under these parameters, calculation of the complete evolution for the spinodal decomposition took around 30 minutes using the IBM Cluster from CRIHAN [22].

Finally, to define the derivative of the free energy density with respect to concentration in the simulations by Eq. (11), we choose a simple double-well function $\tilde{f}=-(c^2-c)+(5/2)(c-1/2)^4$ such that

$$\tilde{f}_c'\equiv\frac{\partial\tilde{f}}{\partial c}=2\left(c-\frac{1}{2}\right)\left(-1+5\left(c-\frac{1}{2}\right)^2\right).$$

4 Numerical results

4.1 Patterns in spinodal decomposition

Results of modeling by Eq. (11) are shown in Fig. 1. These represent snapshots of the evolution of iso-concentration surfaces. It can be seen that randomly distributed concentrations [snapshots (a) and (d) in Fig. 1] create well-known patterns in spinodal decomposition in time [snapshots (b)–(c) and (e)–(f) in Fig. 1].

Comparing snapshots (b) and (e) in Fig. 1, one may conclude that the spinodal decomposition proceeds faster in parabolic evolution exhibiting more developed concentration surfaces

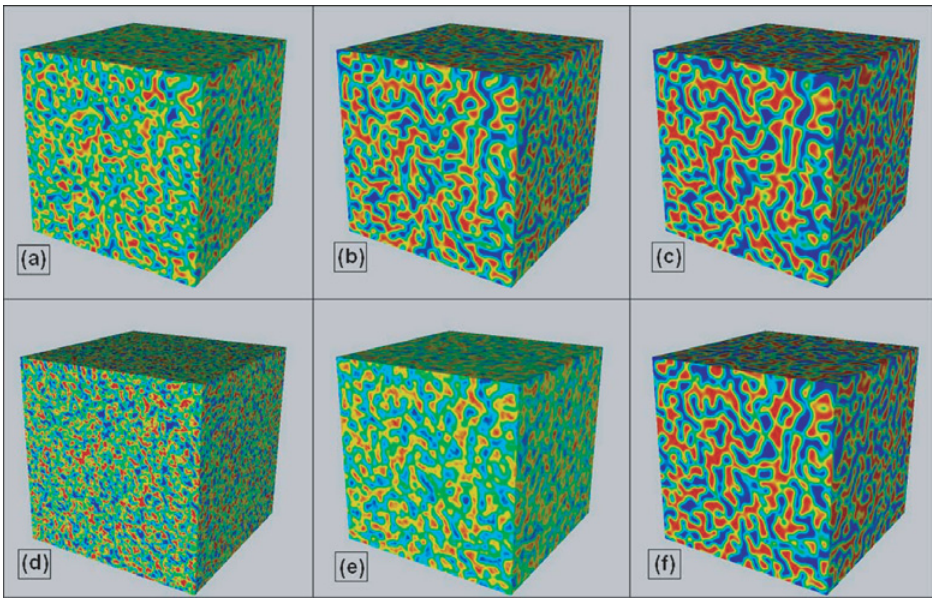


Fig. 1. Evolution of patterns in spinodal decomposition as predicted by numeric modeling: (a)–(c) patterns in Cahn-Hilliard parabolic evolution (Eqs. (7) and (11) with $\tilde{\tau}_D \to 0$); (d)–(f) patterns in hyperbolic evolution (Eqs. (7) and (11) with finite $\tilde{\tau}_D$). (a) and (d): $t = 10^3\tau_D$; (b) and (e): $t = 5{\cdot}10^3\tau_D$; (c) and (f): $t = 10^4\tau_D$.

between separating phases in comparison with the hyperbolic evolution at the same time step ($5{\cdot}10^3\tau_D$). This fact is the result of inertia effects taken into account by the term $\tau_D\partial^2 c/\partial^2 t$ in Eq. (7): spinodal decomposition in hyperbolic evolution develops more slowly than in parabolic evolution with the delay proportional to the local relaxation time τ_D. At the following stages of decomposition [snapshots (c) and (f) in Fig. 1] both evolutions exhibit visually similar patterns. To characterize the parabolic evolution and hyperbolic evolution quantitatively, we further use the structure factor.

4.2 Modeling for the structure factor

Experimentally, a typical structure after spinodal decomposition is also observed as interconnected patterns similar to the model results shown in Fig. 1. Such structure has a characteristic length-scale related to maximal amplification rate of decomposition [23]. Experimental observations using scattering show a broad Bragg-like peak, from which information about the quenched structure during spinodal decomposition and the decomposition rate can be read off. For such measurements, a main quantitative and qualitative characteristic for the intensity of scattering is a structure factor. Therefore, the structure factor can be taken as an important parameter also in modeling for characterization of analyzed phase decomposition and to verify model predictions in comparison with experimental data.

Using Eq. (11), the structure factor is obtained as

$$S_k^n = \frac{\hat{c}_k^n \hat{c}_k^{n*}}{S_0}, \tag{12}$$

where $\hat{c}_k^{n*}$ is the conjugated concentration to $\hat{c}_k^n$ at the time step n. The intensity S_0 for a given grid node (i, j, k) is given by

$$S_0 = \frac{\sum_{ijk}^{N} |\hat{c}_k^n(i, j, k)|^2}{V_0}, \tag{13}$$

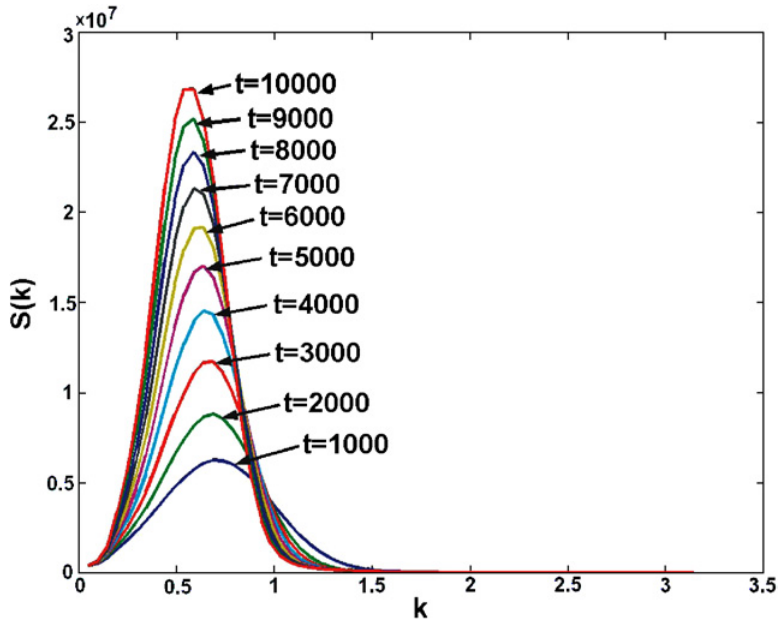


Fig. 2. Evolution of the equal time structure factor in spinodal decomposition as predicted by Eqs. (12)–(13) and Cahn-Hilliard parabolic numerical equation (11) with $\tilde{\tau}_D = 0$ and $\tilde{\epsilon}_c^2 = 1.0$. Numbers at the curves show number of time steps in units of τ_D.

with $\sum_{ijk}^{N} |\hat{c}_k^n(i,j,k)|^2$ the integrated intensity within the volume V_0 of whole computational domain and N-number of nodes along the i-th, j-th, and k-th axes. Hence, at every time step, Eq. (11) yields the structure factor by Eqs. (12)–(13).

Figures 2 and 3 show the structure factor S (scaled by l_D^3) as a function of the wave-number k (scaled by l_D) at the different time t (scaled by τ_D) for parabolic CH-model and the hyperbolic model of decomposition, respectively. Figure 2 shows smooth and monotonic curves for the structure factor $S(k)$ at every time t as predicted by the parabolic CH- and CHC-models. $S(k)$ is also predicted to develop a peak, the height $S_m \equiv S(k_m)$ of which increases with time due to an increase in volume fraction of one of the phases during phase separation. The peak position of S_k is shifted towards smaller values of k, reflecting the increase of the domain growth with time. Contrary to that, the system described by the hyperbolic equation evolves smoothly but with the wave behavior of $S(k)$ especially at large k and small t as is shown in the inset of Fig. 3. At the initial stages of decomposition, $S(k)$ has non-monotonic curves with several peaks. They appear to show that, at a given time, the pattern for one of phases has a volume fraction with different k. As a result, the pattern with the smallest k is selected due to the gradual damping of peaks at larger k. At the following stages of decomposition, the structure factor exhibits only the principal peak of the height $S_m \equiv S(k_m)$. The peak is gradually shifted to smaller wave-numbers k with time (see Fig. 3).

Thus, in the system described by the hyperbolic equation, the wave behavior of $S(t,k)$ indicates that structures with different k exist at a given t. Such behavior means that the system creates various structures during finite time as well as it tries to select the pattern with smallest k. As soon as the system finds a pattern with the wave-number k smaller that some characteristic critical value k_c, it begins to irreversibly grow. Patterns with larger k gradually disappear in such a manner that $S(k)$ exhibits damped oscillations. The finite-time-process of the wave-number selection is the result of taking inertia $\tau_D \partial^2(\delta c)/\partial t^2$ into account by Eq. (4). This process is absent in the system described by the equation of the parabolic type which neglects the inertia. In this case, the domain growth has $k < k_c$ and the structure factor with only principal peak at any time. The behavior with wave developing structure factor $S(t,k)$ and its approaching the scenario predicted by the parabolic CH-model with time can be explained from analytic solution of the problem as is given in the next Section.

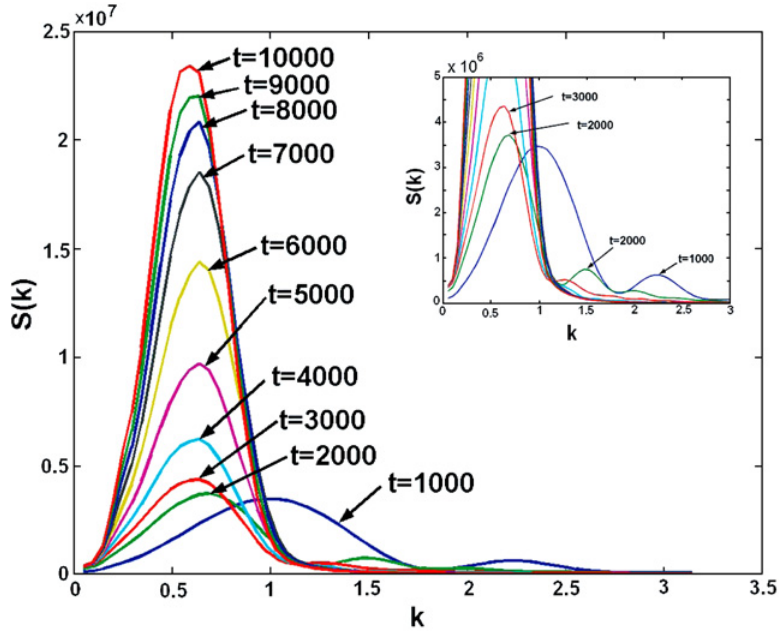


Fig. 3. Evolution of the equal time structure factor $S(k)$ in spinodal decomposition as predicted by Eqs. (12)–(13) and hyperbolic numerical equation (11) with $\tilde{\tau}_D$=1.0 and $\tilde{\epsilon}_c^2$=1.0. Inset shows in details the wave behavior of $S(k)$ at initial stages of decomposition with large wave-numbers k. Numbers at the curves show number of time steps in units of τ_D.

5 Analysis of the structure factor at earliest stages

The semi-implicit Fourier spectral method described in Section 3 allows us to avoid the assumption of linearity and provide description of earliest, intermediate and final stages of decomposition. However, the main qualitative difference in evolution of the structure factor is seen at the initial stages of decomposition. Therefore we focus on the analysis of the initial stages of decomposition described by Eq. (4) (i.e., when the large concentration gradients exist and short periods of time are important) in which one may neglect all terms not linear in c.

Consider the structure factor $S(\mathbf{k}, t)$ which describes the intensity of quasi-elastic scattering observed at time t after quenching from the initial temperature T_i up to the final temperature T_f. The function $S(\mathbf{k}, t)$ can be interpreted as the respective correlation function of the concentrational fluctuations and it is defined as

$$S(\mathbf{k}, t) = \langle \delta c(-\mathbf{k}, t) \delta c(\mathbf{k}, t) \rangle_{T_f}. \tag{14}$$

This expression is the continuous version of Eqs. (12)–(13). To obtain an expression for the time dependent structure factor (14) we linearize Eq. (4) in terms of the concentrational inhomogeneity $\delta c(\mathbf{r}, t) = c(\mathbf{r}, t) - c_0$. This yields

$$\tau_D \frac{\partial^2(\delta c)}{\partial t^2} + \frac{\partial(\delta c)}{\partial t} = M f''_{cc} \nabla^2(\delta c) - M \epsilon_c^2 \nabla^4(\delta c), \tag{15}$$

where $f''_{cc} = \partial^2 f(T, c)/\partial c^2$ is the second derivative of the free energy density $f(T, c)$ with respect to concentration c. Eq. (15) is a linearized partial differential equation of hyperbolic type which is expected to be useful for description of phase decomposition for short periods of time, large concentration gradients, or under deep supercoolings into the spinodal region of a phase diagram.

Using Fourier transforms of $\delta c(-\mathbf{k},t)$ and $\delta c(\mathbf{k},t)$ as

$$\begin{aligned}\delta c(-\mathbf{k},t) &= \int \mathrm{d}\mathbf{r}\exp(i\mathbf{k}\cdot\mathbf{r})\delta c(-\mathbf{r},t),\\ \delta c(\mathbf{k},t) &= \int \mathrm{d}\mathbf{r}\exp(i\mathbf{k}\cdot\mathbf{r})\delta c(\mathbf{r},t),\end{aligned} \tag{16}$$

one can rewrite Eq. (15) for the structure factor (14) in the following form

$$\tau_D\frac{\mathrm{d}^2S(\mathbf{k},t)}{\mathrm{d}t^2}+\frac{\mathrm{d}S(\mathbf{k},t)}{\mathrm{d}t}=-Mk^2\left(f''_{cc}+\epsilon_c^2k^2\right)S(\mathbf{k},t). \tag{17}$$

To solve this equation we multiply by $\exp(i\omega t)$ on both sides, where ω is a frequency of the concentrational inhomogeneity. After some algebra, a general solution of Eq. (17) is given by

$$\int_0^\infty \exp(-i\omega t)S(\mathbf{k},t)\mathrm{d}t=\frac{\tau_D\mathrm{d}S(\mathbf{k},0)/\mathrm{d}t+(1+\tau_Di\omega)S(\mathbf{k},0)}{i\omega-\tau_D\omega^2+Mk^2(f''_{cc}+\epsilon^2k^2)}. \tag{18}$$

Defining the spectral distribution of fluctuations as

$$S(\mathbf{k},\omega)=2Re\left(\int_0^\infty \exp(-i\omega t)S(\mathbf{k},t)\mathrm{d}t\right), \tag{19}$$

one can find from Eq. (18) the spectral distribution of concentrational fluctuations.

Eqs. (18) and (19) describe modeled structure for hyperbolic scenario (with finite τ_D) and parabolic scenario (for instant relaxation with $\tau_D\to 0$). Numerically and experimentally observed structures after quenching in spinodal decomposition exhibit different patterns even at the same time but dependent on the mechanism (parabolic or hyperbolic) of their realization (see Fig. 1). Therefore, Eqs. (18) and (19) can give information about lengthscale of concentrational fluctuations at the very initial stages as well as about the maximal amplification rate selected at long times.

To show explicitly the behavior of the structure factor at the initial stages of decomposition we directly write the solution of Eq. (17) in the following form

$$S(\mathbf{k},t)=s_1e^{p_+t}+s_2e^{p_-t},\qquad p_\pm=\frac{1}{2\tau_D}\left(-1\pm\sqrt{1-4\tau_DM\mathbf{k}^2(f''_{cc}+\epsilon_c^2\mathbf{k}^2)}\right), \tag{20}$$

with s_1 and s_2 arbitrary amplitudes. Because we are interested in the amplification rate giving a faster growing mode, one has to consider only p_+ in solution (20). Consequently, one gets

$$S(\mathbf{k},t)=S(\mathbf{k},0)e^{R(\mathbf{k})t},\qquad R(\mathbf{k})=\frac{\sqrt{1+4\tau_DD\mathbf{k}^2(1-\mathbf{k}^2/\mathbf{k}_c^2)}-1}{2\tau_D}, \tag{21}$$

with $R(\mathbf{k})$ the amplification rate of spinodal decomposition (the spinodal rate), $D=-Mf''_{cc}$ the diffusion coefficient (with $f''_{cc}<0$ in a spinodal region of a phase diagram at a given temperature T), and $\mathbf{k}_c=\sqrt{-f''_{cc}}/\epsilon_c$ the critical wave-number below which perturbations irreversibly grow. Note that $\mathbf{k}_c$ is the same predicted by parabolic and hyperbolic types of equations for spinodal decomposition [14,15].

With the condition

$$1+4\tau_DD\mathbf{k}^2(1-\mathbf{k}^2/\mathbf{k}_c^2)<0 \tag{22}$$

solution (21) becomes complex and describes oscillatory behavior of the perturbations with the exponential decay of their amplitudes during the time of the order of $2\tau_D$. Indeed, Eq. (21) can be re-written as

$$S(\mathbf{k},t)=S(\mathbf{k},0)\exp\left(\frac{i\alpha(\mathbf{k})t}{2\tau_D}\right)\exp\left(-\frac{t}{2\tau_D}\right),\qquad \alpha(\mathbf{k})=\sqrt{1+4\tau_DD\mathbf{k}^2(1-\mathbf{k}^2/\mathbf{k}_c^2)}, \tag{23}$$

in which the oscillations and their exponential decay are explicitly shown by the terms $\exp[i\alpha(\mathbf{k})t/(2\tau_D)]$ and $\exp[-t/(2\tau_D)]$, respectively.

Behavior with damped oscillations predicted by Eq. (23) is absent in the model given by the parabolic CHC-model (Eq. (15) with $\tau_D \to 0$). Obviously, within the limit $\tau_D \to 0$, solution (21) gives the following structure factor

$$S(\mathbf{k},t) = S(\mathbf{k},0)e^{R_{CH}(\mathbf{k})t}, \qquad R_{CH}(\mathbf{k}) = -M\mathbf{k}^2(f''_{cc} + \epsilon_c^2\mathbf{k}^2) = D\mathbf{k}^2(1 - \mathbf{k}^2/\mathbf{k}_c^2), \tag{24}$$

with $R_{CH}(\mathbf{k})$ the amplification rate of spinodal decomposition (the spinodal rate) predicted by the parabolic Cahn-Hilliard equation [17,18].

From solution (21) under condition (22) one can find the critical value $\mathbf{k}_0$ of the wave-number

$$\mathbf{k}_0^2 = \frac{1}{2\varepsilon_c^2}\left(\left((-f''_{cc})^2 + \frac{\varepsilon_c^2}{\tau_D M_c}\right)^{1/2} - f''_{cc}\right), \tag{25}$$

which gives a boundary for modes having oscillations, i.e., a boundary between real and complex values for the amplification rate $R(\mathbf{k})$ given by Eq. (23). The boundary is defined as:

(i) for $\mathbf{k}^2 < \mathbf{k}_0^2$ the relaxation behavior tends in time to those one predicted by the exponential CH-structure factor (24),
(ii) for $\mathbf{k}^2 > \mathbf{k}_0^2$ the relaxation behavior is described by the oscillatory mode (21).

Thus, appearing from the inequality (22), Eq. (25) gives a boundary between rapid oscillating regime and exponential increase of perturbation which can be considered as an ultraviolet cutoff absent in standard parabolic evolution. For large $\mathbf{k} > \mathbf{k}_0$, existing at earliest time periods of decomposition, the structure factor $S(\mathbf{k},t)$ has damped oscillations (23) which disappear in time approaching the form of function (24) predicted by the CH-model. This explains wave behavior of the structure factor for large $\mathbf{k}$ at the very beginning of decomposition and gradual transition in time of the curves with several peaks to the smooth monotonic curve with the unique peak as is shown in Fig. 3.

6 Conclusions

The influence of memory effects on the kinetics of spinodal decomposition has been studied in the example of a hyperbolic model. The model is given by an exponential form of the memory function which contributes to the driving force of decomposition and predicts diffusion and wave modes of atomic propagation.

The hyperbolic model of spinodal decomposition was investigated numerically. A semi-implicit scheme was used to numerically predict evolution of patterns and structure factor $S(k,t)$ in spinodal decomposition. It has been found that the inclusion of memory into the model leads to a delay of the phase separation kinetics due to the appearance of inertia in the hyperbolic model.

Numerical solution of the parabolic equation (with the zero relaxation time τ_D to a steady state atomic diffusion flux) predicts smooth and monotonic curves for the structure factor $S(k,t)$ at every time t as predicted by the parabolic CH- and CHC-models. In this case, the structure factor has the unique peak $S_m \equiv S(k_m)$ which increases to the smaller value of the wave-number k with time.

Numerical solution of the hyperbolic equation (with the finite relaxation time τ_D) predicts $S(k,t)$ given by non-monotonic curves with several peaks at the initial stages of decomposition. In this case, the system creates various structures during the finite time as well as it tries to select the pattern with smallest k (Fig. 3). As soon as the pattern appears with the wave-number k smaller than the critical value k_c [which governs amplification rates (21) and (24)], it begins to irreversibly grow. Patterns with larger k gradually disappear in such a manner that $S(k)$ exhibits damped oscillations. The finite-time-process of the wave-number selection is

the result of taking inertia $\tau_D \partial^2(\delta c)/\partial t^2$ into account by Eq. (4). This process is absent in the system described by the parabolic CH-equation which neglects inertia.

At the following stages of decomposition in the system described by the hyperbolic equation, the structure factor has only one peak of height $S_m \equiv S(k_m)$ which is gradually shifted to smaller wave-numbers k with time. Thus, the hyperbolic scenario gradually tends in time to the dissipative mode predicted by the CH-model. Such behavior with the existence of damped oscillations in $S(k,t)$ at the earlier stages of decomposition and their gradual decreasing in time to follow the smooth monotonic curves with the unique peak, Fig. 3, is explained by an analytical solution obtained.

As a final note, two points might be outlined as a further development and test of the hyperbolic model of spinodal decomposition. *First*, due to the well-known influence of stochastic processes on pattern formation, one can model the effect of noise on patterns in spinodal decomposition. With regard to the present model, the modeling can be generalized in a manner given by the formalism described in Ref. [24] applicable for the hyperbolic model of solute diffusion in a binary system. *Second*, stages with local nonequilibrium relaxation having characteristic time τ_D may proceed during macroscopic time periods that might be fixed in experimental measurements. e.g., as is evaluated for spinodal decomposition in solid state Fe-Cr alloys [25], the relaxation time τ_D has a value of several tens of minutes. It means that the Fe-Cr alloy's sample may be easily quenched and analyzed with respect to the structure factor during very initial stages of decomposition which proceed under local nonequilibrium solute diffusion. A common case could also be seen in a spinodally decomposing glass, where a slow structural relaxation may lead to a temporal non-locality (memory) and long-time-process of the phase separation. Therefore, experimental verification of the existence of the wave behavior in the structure factor during initial stages of decomposition (as is shown in Fig. 3) could be a reasonable test for the hyperbolic model of spinodal decomposition.

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